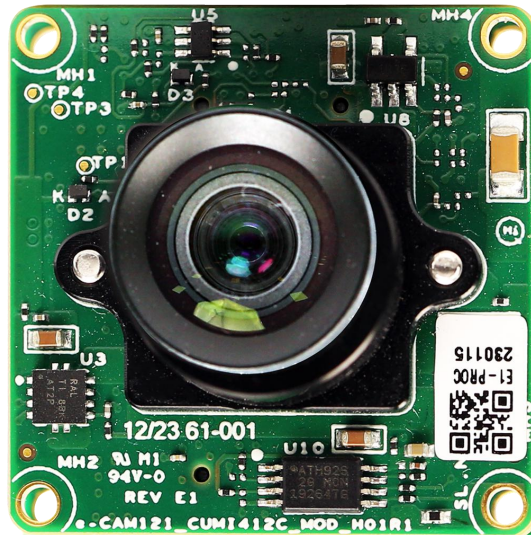




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e-CAM121_CUMI412C_MOD_H01R1



Module Datasheet

Revision 1.0

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1 Introduction

The e-CAM121_CUMI412C_MOD_H01R1 is a low-power, 12 MP camera module from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. This camera module is specifically designed for low-light conditions and is based on IMX412-AACK-D CMOS color image sensor from Sony® STARVIS™ family. It is designed to connect with any application processor that features a MIPI interface.

This document serves as the datasheet for e-CAM121_CUMI412C_MOD_H01R1, providing detailed information on its electrical and mechanical features.

2 Disclaimer

The specifications and features of e-CAM121_CUMI412C_MOD_H01R1 camera board are provided here as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

3 Description

The e-CAM121_CUMI412C_MOD_H01R1 camera module is based on IMX412-AACK-D CMOS image sensor from Sony®. The IMX412-AACK-D is a 1/2.3" optical form-factor CMOS image sensor with an electronic shutter. The e-CAM121_CUMI412C_MOD_H01R1 can stream 4056 x 3040 at 60fps, 4056 x 2288 at 60fps and FHD (2028 x 1112) at 240fps.

The front view of the e-CAM121_CUMI412C_MOD_H01R1 is shown in the following figure.





Figure 1: e-CAM121_CUMI412C_MOD_H01R1 Camera Module

3.1 Camera Modules Features

The features of camera module are as follows:

- 1/2.3" optical form-factor, 12 MP camera module
- Imaging applications: Still capture supported resolutions FHD (2028 x 1112) at 240fps, 4056 x 2288 at 60fps and 4056 x 3040 at 60fps
- Small form factor pluggable camera module with S-mount lens holder
- 30 mm x 30 mm size
- Restriction of Hazardous Substances (RoHS), REACH and FCC compliant
- Dual 20 Pin SMT connector
- Power consumption: 554.4 mW
- Operating temperature: -20C to 60C.

3.2 CMOS Image Sensor Features

The features of CMOS image sensor are as follows:

- IMX412-AACK-D-12 MP RGB RAW 10-bit/12-bit data per pixel CMOS image sensor from SONY®
- 1/2.3" optical form-factor
- Pixel Size: 1.55 μm (H) x 1.55 μm (V)
- Chip size: 7.564 mm (H) x 5.476 mm (V)
- Achieves low power consumption
- Electronic Shutter



3.3 Maximum Frame Rate Supported

The maximum frame rate supported in the e-CAM121_CUMI412C_MOD_H01R1 is listed in the following table.

Resolution	RAW 10-bit	RAW 12-bit
4056 x 3040	60	40
4056 x 2288	60	-
FHD (2028 x 1112)	240	-

Table 1: Maximum Frame Rate Supported

4 Pin Description

The e-CAM121_CUMI412C_MOD_H01R1 has dual row 20-pin SMT connector. The CN1 and CN2 connectors are shown in the following table.

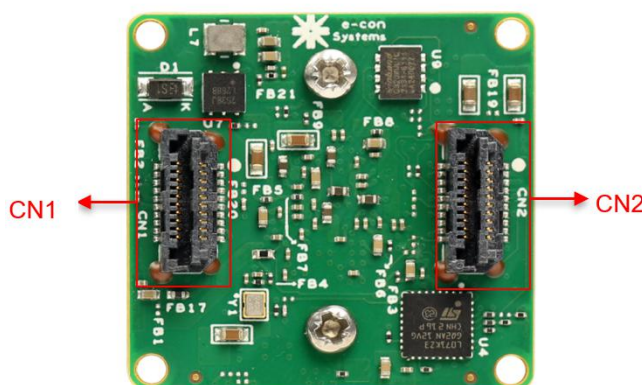


Figure 2: CN1 and CN2 Connectors on e-CAM121_CUMI412C_MOD_H01R1

The CN1 and CN2 pin numbers, signal names, pin types and their description from sensor perspective are listed in the following tables.

CN1 Pin No	Signal Name	Pin Type	Description
1	MIPI_DATA2_N	OUTPUT	MIPI Data Lane 2 Differential Pair -
2	MIPI_DATA0_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
3	MIPI_DATA2_P	OUTPUT	MIPI Data Lane 2 Differential Pair +
4	MIPI_DATA0_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog



7	RSVD	-	No Connection
8	VCC_3P3_CONN	POWER	3.3V Power supply for camera boards
9	RSVD	-	No Connection
10	VCC_3P3_CONN	POWER	3.3V Power supply for camera boards
11	GND	POWER	Ground signal for digital and analog
12	GND	POWER	Ground signal for digital and analog
13	MIPI_DATA3_N	OUTPUT	MIPI Data Lane 3 Differential Pair -
14	MIPI_CLK_N	OUTPUT	MIPI Clock Lane Differential Pair -
15	MIPI_DATA3_P	OUTPUT	MIPI Data Lane 3 Differential Pair +
16	MIPI_CLK_P	OUTPUT	MIPI Clock Lane Differential Pair +
17	GND	POWER	Ground signal for digital and analog
18	RSVD	-	No Connection
19	RSVD	-	No Connection
20	RSVD	-	No Connection

Table 2: CN1 Pin Descriptions

CN2 Pin No	Signal Name	Pin Type	Description
1	RSVD	-	No Connection
2	MIPI_DATA1_N	OUTPUT	MIPI Data Lane 1 Differential Pair -
3	RSVD	-	No Connection
4	MIPI_DATA1_P	OUTPUT	MIPI Data Lane 1 Differential Pair +
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog
7	CAM_I2C_SCL	INPUT	I2C Clock signal
8	RSVD	-	No Connection
9	CAM_I2C_SDA	I/O	I2C Data signal
10	RSVD	-	No Connection
11	nMCU_RESET	INPUT	Active low reset control signal. Contains on-board RC circuits
12	RSVD	-	No Connection
13	PWDN	INPUT	General purpose Input. Used for firmware update of MCU. For normal operation, drive LOW
14	RSVD	-	No Connection
15	RSVD	-	No Connection



16	GND	POWER	Ground signal for digital and analog
17	RSVD	-	No Connection
18	RSVD	-	No Connection
19	RSVD	-	No Connection
20	RSVD	-	No Connection

Table 3: CN2 Pin Description

4.1 Mating Connector Detail

The details of the mating connectors are listed in the following table.

Connector	Description	Manufacturer	Part Number
On board connector (CN1 and CN2)	Board - Board, 20 pin 0.635 mm pitch vertical SMD. Connector mounted on e-CAM121_CUMI412C_MOD_H01R1	Samtec	LSS-110-01-H-DV-A
Mating connector (suggested)	Board - Board, 20 pin 0.635 mm pitch vertical SMD (1 to 2 mating).	Samtec	LSS-110-01-H-DV-A

Table 4: Mating Connector Details

5 Electrical Specification

The electrical specifications of the e-CAM121_CUMI412C_MOD_H01R1 are as follows:

- [Recommended Operating Condition](#)
- [Functional Temperature Range](#)
- [DC Characteristics](#)

Note: The values described in this section are measured in e-con Systems lab, and this can be used as reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.

5.1 Recommended Operating Condition

The recommended operating voltage and current consumption of the e-CAM121_CUMI412C_MOD_H01R1 in active mode is listed in the following table.

Voltage	Current
3.3V	0.168

Table 5: Recommended Operating Condition

5.2 Functional Temperature Range

The functional temperature range of the e-CAM121_CUMI412C_MOD_H01R1 is listed in the following table.

Min	Max
-20C	60C

Table 6: Functional Temperature Range

Note: As the temperature increases, the noise level also increases.

5.3 DC Characteristics

The DC characteristics of the e-CAM121_CUMI412C_MOD_H01R1 general-purpose output signal are listed in the following table.

Symb ol	Parameter	Conditio ns	Min	Typ	Ma x	Uni t
Digital Output Signals						
VOL	Output voltage LOW		0		0.4	
VOH	Output voltage HIGH		1.4		1.8	

Table 7: DC Characteristics of General-Purpose Output Signals

The DC characteristics of the e-CAM121_CUMI412C_MOD_H01R1 reset signal are listed in the following table.

Symb ol	Parameter	Conditio ns	Min	Typ	Max	Unit
Digital Input Signals						
VIL	Input voltage LOW		0		0.8	V
VIH	Input voltage HIGH		1.4		1.8	V
Digital Output Signals						
VOL	Output voltage LOW	IOL= 1.5mA	0		0.4	V

Table 8: DC Characteristics of nRST Signal

Note: e-con Systems recommends the working voltage levels to be typically 1.8V_{DC} and not to reach the maximum limit.



6 Mechanical Specification

The module size is 30 mm x 30 mm and the stack-up height of the board with its mating connector is 6 mm. The height of the S-mount lens holder is 9 mm.

Note: For datasheets of the connectors, the S-mount lens holder, and the module's mechanical drawings in DXF file format, Please write to techsupport@e-consystems.com with your requirement .

The e-CAM121_CUMI412C_MOD_H01R1 board drawing and dimensions are described in the following sections.

6.1 e-CAM121_CUMI412C_MOD_H01R1 Module Mechanical Drawing

The top and bottom views of the e-CAM121_CUMI412C_MOD_H01R1 mechanical drawing with optical orientation is shown in the following figures.

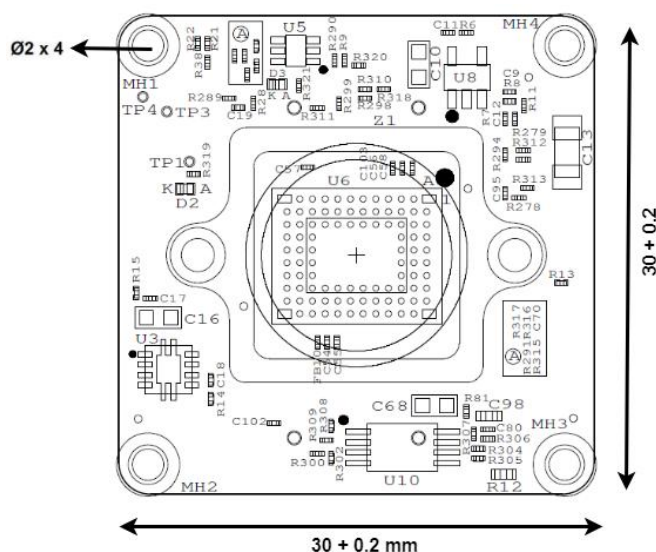


Figure 3: Top View of e-CAM121_CUMI412C_MOD_H01R1 Mechanical Drawing



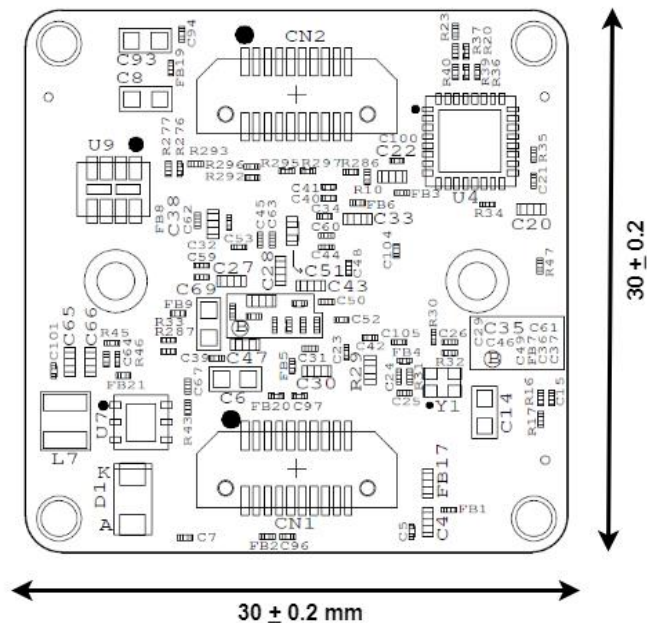


Figure 4: Bottom View of e-CAM121_CUMI412C_MOD_H01R1 Mechanical Drawing

Note: All dimensions are in mm.

6.2 S-Mount Lens Holder Drawing

The mechanical diagram of the S-mount lens holder is shown in the following figure.

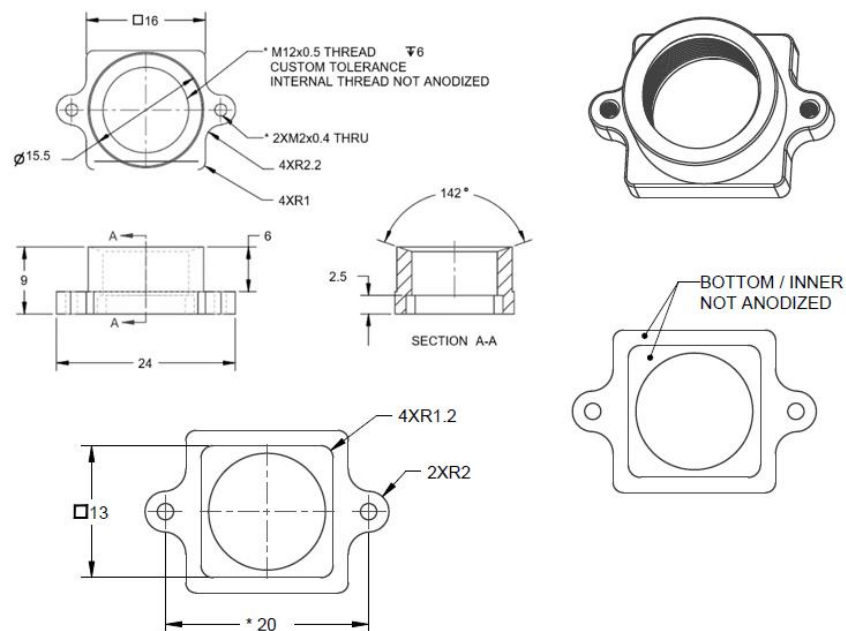


Figure 5: S-Mount Lens Holder Mechanical Diagram

Note: All dimensions are in mm.

Warning: Please ensure that the lens used does not come into contact with the sensor when attempting to use a different lens on the e-CAM121_CUMI412C_MOD_H01R1. Some lenses might have protruded lens filters that could damage the sensor. Check the compatibility of the lens with the holder concerning the tolerance of the lens holder and lens.

6.3 Mechanical Part Details

The mechanical accessories for the e-CAM121_CUMI412C_MOD_H01R1 camera board are listed in the following table.

Part	Specification	Quantity
Lens Holder	Standard S-Mount M12 x P0.5, Aluminum Lens Holder	1
Lens Holder Screw	1.6 mm diameter self-tapping screws	2
Module connector (LSS-110-01-H-DV-A)	Board - Board, 20 pin 0.635 mm pitch vertical SMD.	2

Table 9: Mechanical Parts Details

Support

Contact Us

If you need any support on e-CAM121_CUMI412C_MOD_H01R1 product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>



Revision History

Rev	Date	Description	Author
1.0	13-June-2023	Initial Draft	Camera Team

